

Yoshitaka Inoue

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Machine Learning for Molecular Medicine and Therapeutic Response

University of Minnesota | Computer Science and Engineering

National Institutes of Health | NLM / NCI

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Google Scholar | ORCID | GitHub

Research interests: Machine learning for molecular medicine; graph neural networks; drug response and perturbation modeling; single-cell analysis; intervention-aware representation learning; LLM-based biomedical reasoning.

Education

	University of Minnesota, Twin Cities <i>PhD Candidate, Computer Science and Engineering</i>
2021–present	Graph representation learning, drug-response prediction, single-cell modeling, and biomedical LLM reasoning. Advisor: Prof. Rui Kuang; Co-advisor: Dr. Augustin Luna (NIH). Expected September 2027.
	Nara Institute of Science and Technology <i>M.Eng., Information Science</i>
2017–2019	MS/MS-based small-molecule prediction and annotation with UC Davis Genome Center/DataLab. Advisor: Prof. Shigehiko Kanaya.
	University of California, Davis <i>Visiting Scholar, Genome Center and DataLab</i>
Summer 2018	MS/MS-based compound prediction and metabolomics. Advisors: Prof. Oliver Fiehn and Tobias Kind.
	Kumamoto University <i>B.Sc., Biological Science</i>
2012–2017	Experimental analysis of seagrass metabolism using molecular biology methods. Advisor: Prof. Susumu Takio.

Research and Professional Experience

	National Library of Medicine / National Cancer Institute, NIH <i>Pre-doctoral Fellow</i>
2024–present	Graph neural networks and multi-agent AI for drug response, drug-target reasoning, and biomedical evidence integration across pharmacogenomic, single-cell, screening, cell-line, and patient data.
	Google Summer of Code <i>Contributor (2022); Mentor (2023–2024)</i>
2022–2024	Built a drug-response prediction pipeline; mentored PyTorch Geometric dataset and multi-tool LLM aggregation projects.
	FreakOut, Inc. <i>Machine Learning Engineer</i>
2019–2021	Applied machine learning to click-through-rate prediction in Tokyo, Japan.

Selected Publications

- Inoue, Y., et al. DrugAgent: Reliable Multi-Agent Aggregation under Conflicting Biomedical Evidence.**
Scientific Reports, 2026, under revision
- Inoue, Y., et al. drGT: Attention-Guided Gene Assessment of Drug Response Utilizing a Drug-Cell-Gene Heterogeneous Network.**
BMC Bioinformatics, 2026
- Inoue, Y., et al. Drug Discovery in the Era of Artificial Intelligence: From Target Identification to Clinical Trials.**
Health Data Science, 2026, accepted
- Hoshi-Kadonosawa, Y., et al. (incl. Inoue, Y.) **TRPM4 Expression as a Predictive Biomarker and a Mechanistic Driver of Acetalax Activity in Prostate Cancer.**
Molecular Cancer Therapeutics, 2026
- Elloumi, F., et al. (incl. Inoue, Y.) **CellMiner cross-database (CellMinerCDB) version 2.2 for explorations of patient-derived cancer cell line pharmacogenomics.**
Nucleic Acids Research, 2026
- Taniyama, D., et al. (incl. Inoue, Y.) **Spontaneous dynamic gene expression of Schlafen 11 (SLFN11) demonstrated by single-cell RNA sequencing in clonal cancer cell lines.**
Oncogene, 2026, submitted

Selected Conference Papers and Preprint

- Inoue, Y., et al. DrugAgent: Reliable Multi-Agent Aggregation under Conflicting Biomedical Evidence.**
ICML Workshop on Foundation Models for the Life Sciences (FM4LS), 2026
- Inoue, Y., Fu, T., & Luna, A. GraphPINE: Graph Importance Propagation for Interpretable Drug Response Prediction.**
ICLR Workshop on Machine Learning for Genomics Explorations (MLGenX), 2025
- Inoue, Y., et al. DrugAgent: Explainable Drug Repurposing Agent with Large Language Model-based Reasoning.**
ICLR Workshop on Machine Learning for Genomics Explorations (MLGenX), 2025
- Inoue, Y., Kulman, E., & Kuang, R. BiGCN: Leveraging Cell and Gene Similarities for Single-cell Transcriptome Imputation with Bi-Graph Convolutional Networks.**
bioRxiv, 2024

Selected Presentations		
International Conference on Intelligent Systems for Molecular Biology (ISMB)		
Two accepted oral presentations: <i>drGT</i> in <i>NetBio</i> and <i>DrugAgent</i> in <i>Text Mining</i> ; both also presented at the Student Council Symposium.	2026	Washington, DC
26th Annual CCR-FYI Colloquium		
<i>drGT</i> oral presentation; selected as one of four Outstanding Postgraduate Fellow Finalists	2026	Rockville, MD
NIH Artificial Intelligence Symposium		
<i>drGT</i> : attention-guided gene assessment for drug-response modeling	2025	Rockville, MD

Honors and Fellowships	
• NIH Pre-Doctoral Research Fellowship, 2024–present. • Outstanding Postgraduate Fellow Finalist, 26th Annual CCR-FYI Colloquium, 2026; selected as 1 of 4 finalists. • Research Visit Scholarships, NAIST and JASSO support for UC Davis research visits, 2018–2019.	
Academic Service and Memberships	
Journal reviewing	IEEE/ACM Transactions on Computational Biology and Bioinformatics (2025–2026); Nature Computational Science (co-reviewer, 2026).
Conference reviewing	ACM BCB (2023); ICML FM4LS (2026); ICML GenBio (2026); IEEE BIBM (2026); ACM SIGSPATIAL (2026).
Memberships	International Society for Computational Biology (ISCB); American Association for Cancer Research (AACR); American Society of Clinical Oncology (ASCO).
Technical Skills	
ML / AI	Graph neural networks; graph representation learning; LLM-based agents; interpretable machine learning.
Computational biology	Pharmacogenomics; biological networks; single-cell RNA-seq; multi-omics analysis.
Programming	Python; PyTorch; PyTorch Geometric; scikit-learn; pandas; NumPy.
Infrastructure	Git; Linux; Jupyter; Slurm; GPU computing; HPC; LangChain.